SANJEEVA KUMAR MD

Full-Stack Developer & Video Editor

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EXPERIENCE & KEY PROJECTS

Full-Stack Developer & Architect - UCHN (Unified Care & Health Network)

Healthcare Coordination Platform & Progressive Web App (PWA)

Completed Prototype | Tech: Next.js 14, React 18, TypeScript, PostgreSQL, Prisma, Socket.IO, Tailwind CSS

Engineered an end-to-end healthcare coordination system unifying clinics, doctors, pharmacies, and patients.

- Architected clinic discovery and doctor specialty lookup with slot collision prevention algorithms.
- Designed a 5-stage emergency SOS response pipeline (Pending -> Assigned -> Dispatched -> In-Progress -> Resolved).
- Integrated real-time waiting queue telemetry using Socket.IO to decrease clinic lobby wait times.
- Implemented pharmacy stock verification with transactional ACID boundaries in Prisma ORM.
- Secured application with Role-Based Access Control (RBAC), JWT authentication, and PWA offline capability.

AI & Computer Vision Developer - Hand Gesture Recognition AI

Computer Vision & Contactless Human-Computer Interaction

Working Prototype | Tech: Python, OpenCV, MediaPipe, NumPy

- Engineered real-time 21-point hand landmark extraction at 30+ FPS under varying lighting conditions.
- Classified finger articulation states to execute digital commands without physical contact.

Embedded Systems & Software - IoT Prototyping & Scheduling Systems

Microcontroller Telemetry & Manufacturing Dispatch Optimization

Prototypes | Tech: ESP32, Arduino Uno, C/C++, Sensors (I2C/SPI), Python, React

- Engineered IoT environmental stations using ESP32/Arduino, streaming sensor telemetry over Wi-Fi/WebSockets.
- Developed dynamic production scheduling engine with FIFO/SPT rules minimizing workstation idle time.

EDUCATION

B.Tech in Information Technology (AR/VR)

Alliance University

Bengaluru, Karnataka, India | Coursework: Software Systems, Real-Time Graphics, Data Structures & IoT

KEY HIGHLIGHTS & CORE CAPABILITIES

Full-Stack Architecture
Built scalable web architectures with Next.js, Prisma, and real-time Socket.IO event coordination.

Hardware & IoT Telemetry
Interfaced Arduino and ESP32 with multi-sensor arrays, serial protocols, and live

Computer Vision & AI
Developed real-time hand tracking and dynamic algorithmic schedulers for manufacturing

DaVinci Resolve & Video Editing
Intermediate editor skilled in narrative pacing, match cutting, and visual storytelli

SKILLS

Next.js | React | TypeScript | Tailwind CSS | PostgreSQL | Prisma ORM | Socket.IO | REST APIs | Python | Java | C | Arduino | ESP32 | Sensors (I2C/SPI) | DaVinci Resolve | Video Editing | AI-Assisted Workflows